

HR Attrition Analytics Dashboard

Analyzing employee attrition patterns using SQL and Power BI to identify key drivers of turnover and deliver actionable retention insights through interactive dashboards.



Project Objectives



Analyze Trends

Identify attrition patterns across departments and job roles.



Interactive Dashboard

Build a Power BI dashboard to support HR decision-making.



Root Cause Study

Examine salary, overtime, satisfaction, and promotions as attrition drivers.



Retention Insights

Provide data-driven strategies to reduce employee turnover.

Tools & Technologies



MySQL

Data analysis and querying

Power BI

Dashboard creation and data visualization

DAX

KPI measures and calculations

Excel / CSV

IBM HR Analytics Dataset

Key KPIs at a Glance

1,470

Total Employees

237

Employees Left

16.12%

Attrition Rate

\$6,503

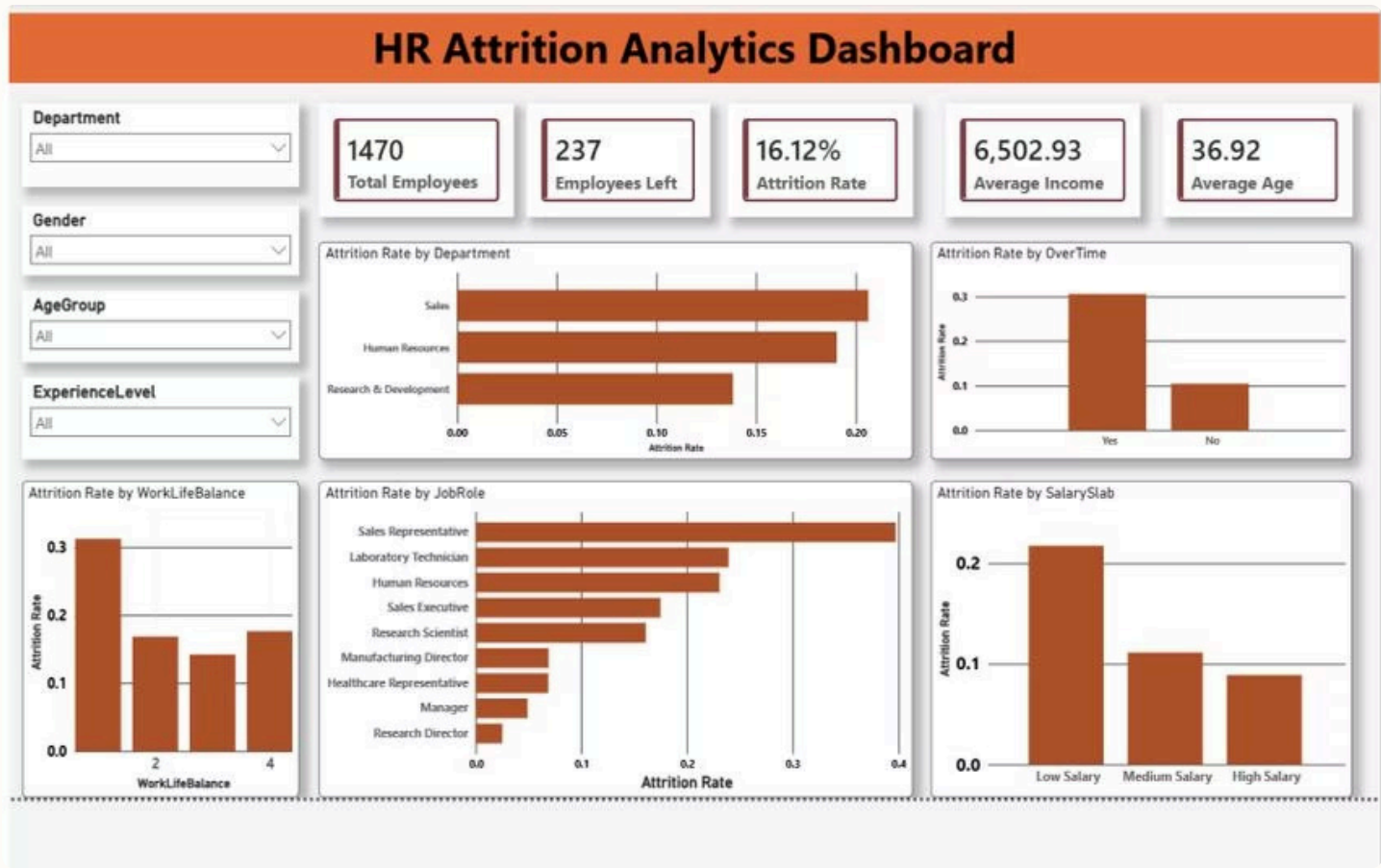
Avg. Income

36.92

Average Age

Based on the IBM HR Analytics Dataset — 237 out of 1,470 employees left, yielding an overall attrition rate of **16.12%**.

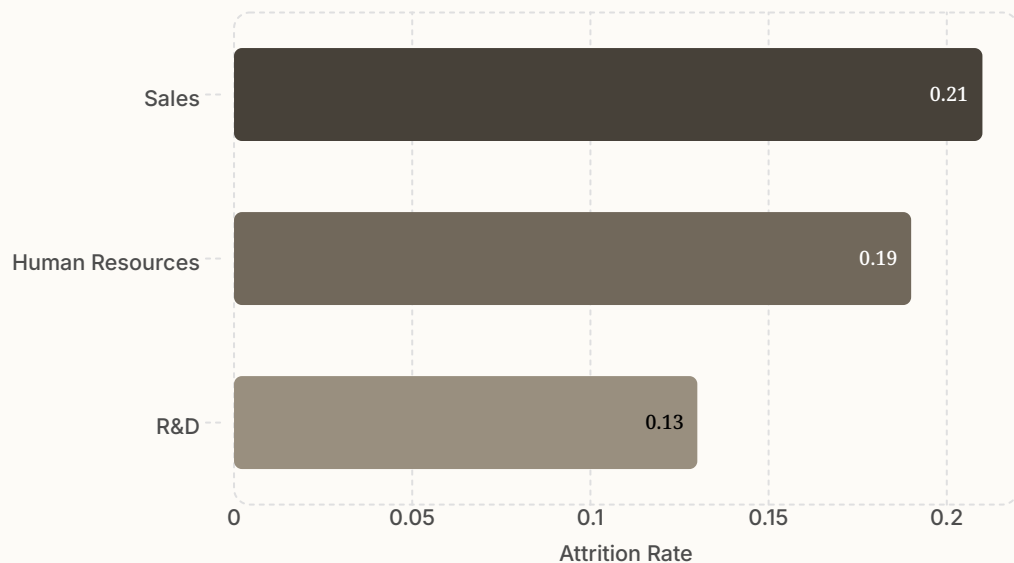
Dashboard — Page 1



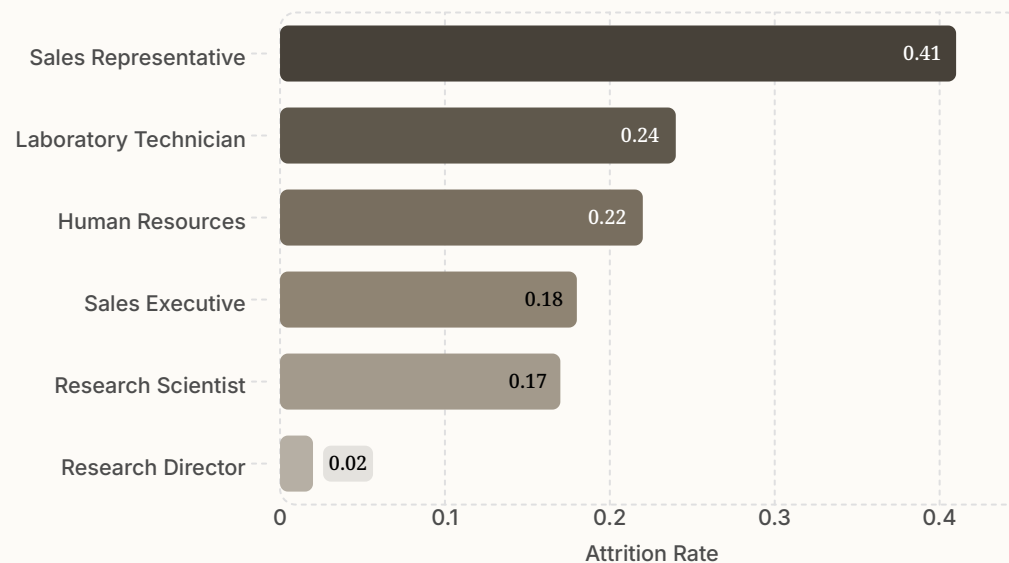
Page 1 covers department, job role, overtime, work-life balance, and salary slab attrition breakdowns with interactive slicers for Department, Gender, Age Group, and Experience Level.

Attrition by Department & Job Role

Department

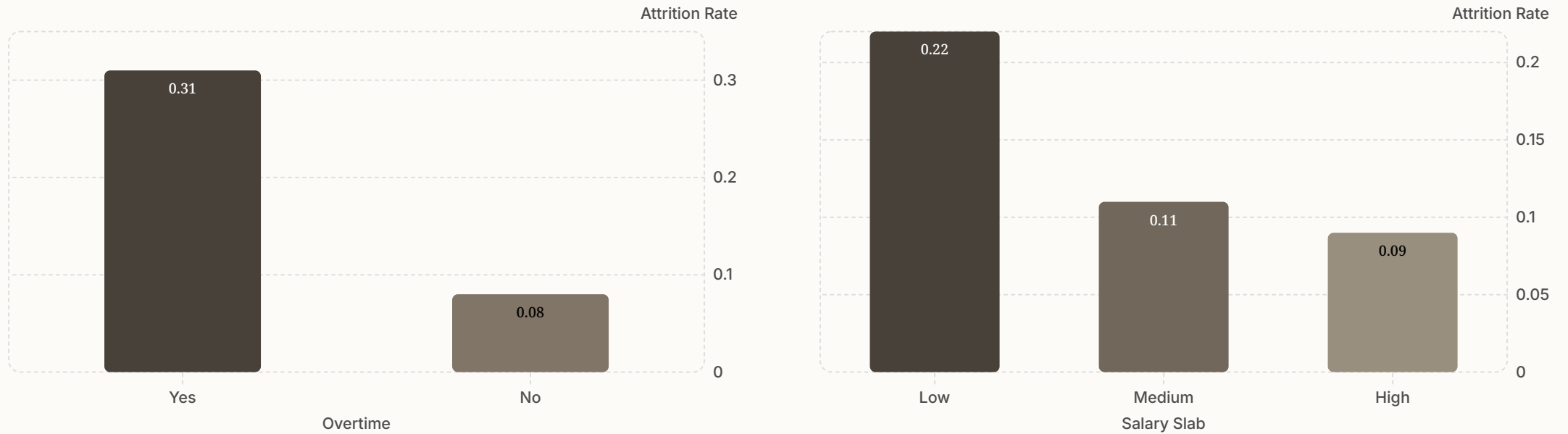


Job Role



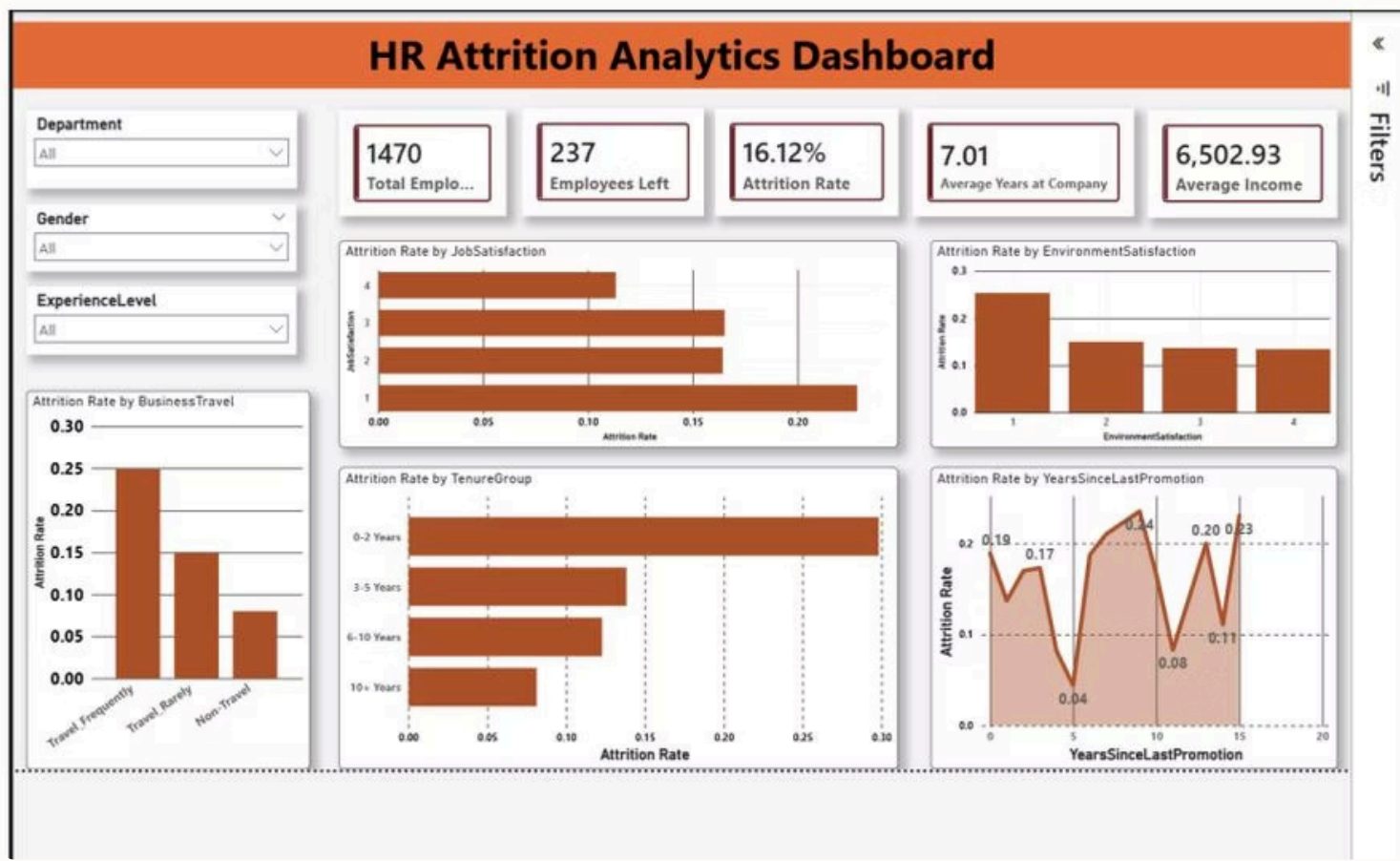
Sales Representatives lead attrition at 41%, nearly double the next highest role. The Sales department overall tops department-level attrition at 21%.

Overtime & Salary Drive Attrition



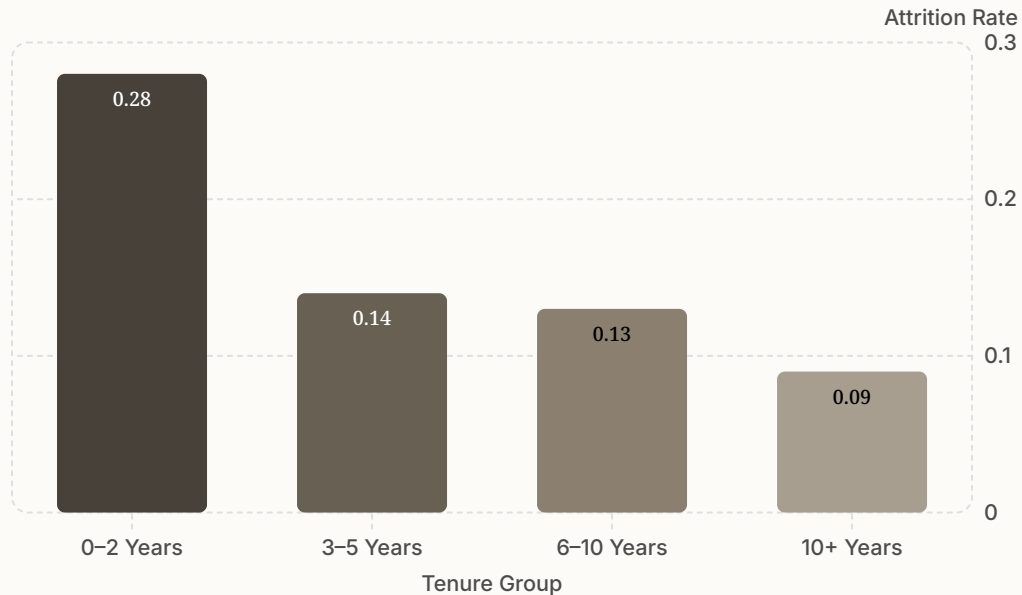
Employees working overtime are ~4× **more likely** to leave (31% vs. 8%). Low-salary employees also show more than double the attrition rate of high-salary peers.

Dashboard — Page 2



Page 2 covers job satisfaction, environment satisfaction, business travel, tenure group, and years since last promotion — enabling root-cause behavioral analysis.

Satisfaction, Travel & Tenure Patterns



Job Satisfaction

Level 1 → **23%** attrition
vs. Level 4 → **11%**

Environment Satisfaction

Level 1 → **24%** attrition
vs. Level 4 → **12%**

Business Travel

Frequent travelers → **25%** vs. Non-travelers → **8%**

New hires (0-2 years) are the highest-risk group at 28%. Low satisfaction and frequent travel compound attrition risk significantly.

Conclusion & Future Enhancements

The dashboard successfully surfaces critical attrition drivers — enabling data-driven HR retention strategies.



Predictive Modeling

Machine Learning for attrition forecasting



Real-Time Integration

Live dashboard data feeds



Sentiment Analysis

Employee feedback and sentiment tracking



Advanced Drill-Through

Deeper forecasting and drill-through features

